

Tandberg TDV-2200 Comprehensive Diagnostic Breakpoint Table

Common Error Handling Points

PRIMARY BREAKPOINT - Universal Error Handler

Address	Function	Validate	Description
0x0695	Error Handler Entry	Register A = Error Code	All diagnostic failures converge here
		Carry Flag = 1	Error code in A register before stack operations
		Stack intact	Called via JMP from all test failures

PRIMARY BREAKPOINT - Universal Success Handler

Address	Function	Validate	Description
0x068E	Success Handler Entry	Carry Flag = 0	All diagnostic successes converge here
		Stack intact	Called via JMP from all test successes

Main Diagnostic Flow Breakpoints

Address	Function	Validate	Description
0x023D	Diagnostic Entry Decision	0x5E0C checksum	If checksum bad, calls diagnostics
		Carry/Zero flags	Decision point for diagnostic execution
0x0240	Post-Diagnostic Initialization	Return from 0x090A	System initialization after diagnostics
0x0244	Error Display Check	Carry Flag	If set, will display error message
0x024A	Error Message Display	HL = 0x0283	Points to "ERROR NO.:" string

Individual Test Breakpoints

Memory Test

Address	Function	Validate	Description
0x0679	Memory Test Call	HL=0x5800, DE=0x5FF0	Memory range being tested
0x067C	Post-Memory Test	Carry Flag	Memory test result

Character Generator RAM Test

Address	Function	Validate	Description
0x0681	CNC 0x0779	Carry Flag = 0	Only calls if memory test passed
0x07C3	Error 0x0F Path	A = 0x0F	Memory write/read mismatch
0x07C8	Error 0x11 Path	A = 0x11	I/O port read/write mismatch
0x07C0	Success Path	Jump to 0x068E	Character generator test passed

Serial Communication Test

Address	Function	Validate	Description
0x0684	CNC 0x0894	Carry Flag = 0	Only calls if char gen test passed
0x08C3	First Pattern Check	A = received data	Should be 0xAA
0x08DE	Second Pattern Check	A = received data	Should be 0xAA
0x08F1	Timeout Error	A = 0x16	No response received
0x08EC	Buffer Error	A = 0x15	Buffer flag error
0x08C5	Pattern Error 1	A = 0x17	First pattern mismatch
0x08E0	Pattern Error 2	A = 0x18	Second pattern mismatch

Extended Memory Range Test

Address	Function	Validate	Description
0x0687	CNC 0x06DE	Carry Flag = 0	Only calls if serial test passed
0x06E7	Early Exit Check	A = 0x1A	Immediate failure condition
0x0701	Range Error	A = calculated	Failed memory address encoded

Keyboard Controller Test

Address	Function	Validate	Description
0x068A	CNC 0x069D	Carry Flag = 0	Only calls if memory test passed
0x06B8	Response Check	A = 0x5E01 value	Should be 0xAA
0x06BD	No Response Error	A = 0x19	Did not receive 0xAA

Error Code Reference Table

Error Code	Test Module	Meaning	Breakpoint
0x0F	Character Generator	Memory write/read mismatch	0x07C3
0x11	Character Generator	I/O port read/write mismatch	0x07C8
0x15	Serial Communication	Buffer flag error	0x08EC
0x16	Serial Communication	Timeout - no response	0x08F1
0x17	Serial Communication	First pattern mismatch	0x08C5
0x18	Serial Communication	Second pattern mismatch	0x08E0
0x19	Keyboard Controller	No 0xAA response received	0x06BD
0x1A	Extended Memory	Immediate test failure	0x06E7
Calculated	Extended Memory	Failed address encoded	0x0701

Memory Locations to Monitor

Address	Purpose	Monitor For
0x5E01	Keyboard response accumulator	Should become 0xAA during keyboard test
0x5E07	Serial receive buffer flag	Used in serial communication test
0x5E0C	System state checksum	Determines if diagnostics run
0x5C2A	Serial function pointer	Modified during serial test
0x5E05	Saved serial pointer	Backup of 0x5C2A

Recommended Debugging Strategy

Single Breakpoint Approach

```
assembly

# Set ONE breakpoint at the universal error handler
break 0x0695

# When hit, examine:
# Register A = Error code (see table above)
# This tells you exactly which test failed and why
```

Comprehensive Analysis

```
assembly
```

```
# Primary decision points
break 0x023D    # Diagnostic entry decision
break 0x0244    # Error display check
break 0x0695    # Universal error handler
break 0x068E    # Universal success handler

# When any breakpoint hits:
# 1. Check carry flag
# 2. Check register A for error codes
# 3. Check relevant memory locations
```

Test-Specific Deep Dive

```
assembly
```

```
# If you know which test is failing, set specific breakpoints:

# For Serial Communication issues:
break 0x08C3    # First pattern check
break 0x08DE    # Second pattern check
info registers  # Check A register for received pattern

# For Character Generator issues:
break 0x07C3    # Memory error path
break 0x07C8    # I/O error path

# For Keyboard Controller issues:
break 0x06B8    # Response validation
examine 0x5E01  # Check accumulated response
```

Quick Validation Commands

At 0x0695 (Error Handler):

```
gdb

info registers          # Check A register for error code
print/x $a              # Error code in hex
print "Error code: 0x%02x", $a  # Formatted error code
```

At 0x068E (Success Handler):

`gdb`

```
print "Test passed"           # All tests successful
info registers                 # Verify carry flag = 0
```

Memory Examination:

`gdb`

```
examine/1bx 0x5E01            # Keyboard response
examine/1bx 0x5E07            # Serial buffer flag
examine/2bx 0x5E0C            # System checksum
```

Error Code Quick Reference

0x0F/0x11: Character Generator Hardware Issue

0x15-0x18: Serial Communication Problems

0x19: Keyboard Controller Not Responding

0x1A/Calculated: Memory Range Issues

Use **breakpoint 0x0695** as your primary diagnostic point - it catches ALL failures with the error code in register A!